



### Boosting solar cell efficiency

A new device design that channels heat into light could potentially boost solar cell efficiency by 80%.  
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### The cells that make up a heart

For the first time, we know the full spectrum of cells that come together to make a heart.  
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# RNA-BASED THERAPIES

After the proliferation of DNA-based gene therapies, RNA-based therapies have emerged displaying a substantial potential to cure rare and common diseases alike

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NA-based therapies have been fairly common in the medical community for decades now.

However, RNA-based therapies are probably something that you haven't heard enough of yet, simply because of their novelty. Unlike the conventional gene therapy which imparts new DNA to cells, RNA-based therapies essentially modify or provide RNA (ribonucleic acid) to the patients' cells. Gene-editing technologies like CRISPR have seen their fair share of popularity, however new advancements in RNA

therapeutic drugs will attempt to address some of the former's serious limitations. This new therapy has the potential to treat an array of rare and common diseases including cardiovascular diseases, neurological diseases, haemophilia, cancer, Huntington's disease and many more.

### RNA-BASED THERAPIES

While RNA therapy and DNA (or gene) therapy are often bunched together, they are actually quite distinct classes of treatment. A theory by Francis Crick relays that genetic information is passed from DNA to RNA and finally to proteins. Then, a type of RNA known as messenger RNA (mRNA) is translated into protein

using a biological code. This converts three nucleic acids to one amino acid.

However, we now know, in addition to Crick's theory of RNA's role, which is limited to carrying genetic information, they are also animated, powerful commanders in human cells' functionalism. Researchers have deciphered types of RNA that are responsible for important activities such as coding and organising protein creation. In addition to that, they also modify the transmission of instructions for DNA, destroy other RNA and prevent rearrangement of genomes.

RNA therapies focus directly on what is referred to as the software of the cell which gives way to be able to intercept a patient's abnormality before it gets translated into functioning or non-functioning proteins. It prevents the production of proteins encoded by a disease.

Both RNA and DNA-based therapies target cells, however, RNA therapy focuses on post-transcriptional therapy that specifically targets mRNA or uses mRNA in the cytoplasm. DNA therapies act in the nucleus and even manipulate the genome, whereas RNA therapies have direct biological targets that